Band-1 Computational Experiment: Brocard's Problem (n!+1 = k^2)

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note is a **finite exact-search probe** (collapse list). It verifies the three known Brown-number solutions and searches $n = 1..N$ with Python arbitrary-precision integers. It does **not** prove that no further solutions exist for all n, does **not** claim the Brocard–Ramanujan problem solved, and does **not** claim Clay Millennium status or Theory of Everything progress. **Problem status: OPEN.**

Abstract

We study Brocard's equation

$$n! + 1 = k^2$$

for positive integers n, k. On this box we:

1. Exactly verify the classical hits $(n, k) = (4, 5)$, $(5, 11)$, $(7, 71)$.
2. Exhaustively test $n = 1, \ldots, N$ with $N = \mathbf{1000}$ using Python `int` and `math.isqrt`, recording every n for which $n! + 1$ is a perfect square.
3. Ledger literature status: the Brocard–Ramanujan problem remains **OPEN**; computational literature reports no further solutions through $n \leq 10^{15}$ (not re-proved here).

Box result: hits only at $n = 4, 5, 7$; **no extras** in 1..1000.

Problem statement

Brocard–Ramanujan problem (Brocard 1876/1885; Ramanujan 1913). Find all integers $n \geq 1$ such that $n! + 1$ is a perfect square. The known solutions (Brown numbers) are

$$4! + 1 = 5^2, \qquad 5! + 1 = 11^2, \qquad 7! + 1 = 71^2.$$

Erdős conjectured there are no others. Overholt (1993) showed that the *abc* conjecture implies only finitely many solutions — still conditional, since *abc* is open.

This experiment. Exact factorial-and-isqrt search on $1..N$ for box-feasible N, plus a literature ledger. No infinitary or conditional-*abc* claim is asserted as a box theorem.

Literature ledger (not re-proved on box)

Item	Status	Note
Brocard–Ramanujan equation	**OPEN**	Only three known solutions; existence of further n unknown
Known solutions (Brown numbers)	Classical	$(4,5)$, $(5,11)$, $(7,71)$
Erdős conjecture	Conjecture	No further solutions (unproved)
Overholt 1993	Conditional	$abc \Rightarrow$ finitely many solutions
Computational searches (lit)	Literature bound	No further solutions for $n \leq 10^{15}$ (Wikipedia / QR-sieve lineage; Berndt–Galway; Matson $\sim 10^{12}$; later $\sim 10^{15}$)
Guy UPNT D25	Survey	Unsolved Problems in Number Theory, 3rd ed.

Critical distinction. A search to $N = 1000$ (or even literature 10^{15}) is a **finite certificate**, not a proof that the solution set is exactly $\{4, 5, 7\}$.

Method (box)

1. Maintain running factorial $f \leftarrow f \cdot n$ with Python `int` (unbounded).
2. For each $n = 1..N$, set $m = f + 1$, compute $r = \text{isqrt}(m)$, accept iff $r^2 = m$.
3. Unit tests: verify known triples; assert nearby non-solutions $n \in \{1, 2, 3, 6, 8, 9, 10\}$; assert search to 20 yields exactly $\{4, 5, 7\}$.

Complexity. Dominated by multiplying and taking integer square roots of integers with $O(N \log N)$ bits; for $N = 1000$, 1000! has 2568 decimal digits — trivial on the box ($\sim 0.01\,\text{s}$).

Results (this box, 2026-09-11)

Quantity	Value
N searched	1000
Solutions found (n)	$4, 5, 7$
Extra solutions in range	**none**
Verified pairs (n, k)	$(4,5)$, $(5,11)$, $(7,71)$
digits($N!$)	2568
Unit tests	**PASS**
Wall time	$\approx 0.013\,\text{s}$
Literature status	**OPEN**
Lit search bound (cited, not re-run)	$n \leq 10^{15}$

Explicit checks

$$4! + 1 = 24 + 1 = 25 = 5^2$$

$$5! + 1 = 120 + 1 = 121 = 11^2$$

$$7! + 1 = 5040 + 1 = 5041 = 71^2$$

Honesty verdict

Finite verification only. Exact search to $N = 1000$ recovers precisely the three known Brown numbers and finds no others in range. Literature computational bounds extend far beyond this box but likewise do not close the problem. **Brocard's problem remains OPEN.** This note does **not** claim Brocard solved, does **not** claim no further solutions for all n, and does **not** claim Clay / ToE.

Reproducibility

```
cd /workspace/experiments/band1/brocard
python3 brocard_probe.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `README.md`.

References (selected)

1. H. Brocard, Question 1532, *Nouv. Corresp. Math.* **2** (1876); *Nouv. Ann. Math.* **4** (1885).
2. S. Ramanujan, Question 469, *J. Indian Math. Soc.* **5** (1913), 59.
3. M. Overholt, The Diophantine equation $n! + 1 = m^2$, *Bull. London Math. Soc.* **25** (1993), 104.
4. R.K. Guy, *Unsolved Problems in Number Theory*, 3rd ed., Springer, 2004, §D25.
5. Wikipedia, *Brocard's problem* (accessed 2026-09-11): searches report no further solutions for $n \leq 10^{15}$.
6. B.C. Berndt & W.F. Galway, On the Brocard–Ramanujan Diophantine equation $n! + 1 = m^2$, *Ramanujan J.* **4** (2000), 41–42 (QR-sieve context).